

MATH 211: CALCULUS III

Spring 2018–19 • First Exam • 11 March 2019

1. [5 pts each] In each part, choose the correct answer.

i. $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^4 + 3 \cos(y^4)}{x^2 + y^2} = \underline{\hspace{2cm}}.$

- (a) 0 (b) 2 (c) 3 (d) it does not exist (e) 4

ii. The linearization of the function $f(x, y) = x \cos(xy) + \ln(x + y)$ at the point $(1, 0)$ is $\underline{\hspace{2cm}}.$

- (a) $2x + y + 1$ (b) $2x + y - 1$ (c) $2x - y + 3$
(d) $x + 2y - 1$ (e) $x + 2y + 1$

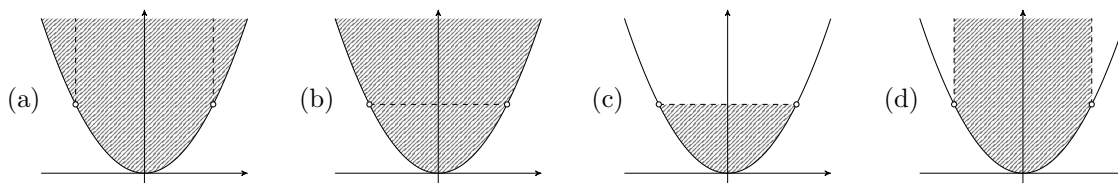
iii. If the equation $e^w = xy^w + 1$ defines w as a differentiable function of x and y , then $\frac{\partial w}{\partial x} = \underline{\hspace{2cm}}.$

- (a) $\frac{-y^w}{e^w - xy^{w-1} \ln y}$ (b) $\frac{e^w - y^w}{e^w - xwy^{w-1}}$ (c) $y^w + xy^w \ln y$
(d) $\frac{y^w}{e^w - xwy^{w-1}}$ (e) $\frac{y^w}{e^w - xy^w \ln y}$

iv. If $f(x, y) = \sqrt{1 + xy^3}$, then $f_{xx} = \underline{\hspace{2cm}}.$

- (a) $\frac{1}{4(1 + xy^3)^{3/2}}$ (b) $\frac{y^6}{4(1 + xy^3)^{3/2}}$ (c) $\frac{-y^6}{4(1 + xy^3)^{3/2}}$
(d) $\frac{3y^6}{4(1 + xy^3)^{3/2}}$ (e) $\frac{-3y^6}{4(1 + xy^3)^{3/2}}$

v. Which of the following sketches best describes the domain of the function $f(x, y) = \frac{\sqrt{y - x^2}}{4 - x^2}$?



2. [15 pts] Let $u = \sqrt{x^2 + y^2}$, where $x = r + s \cos t$ and $y = s + r \sin t$. Find $\frac{\partial u}{\partial r}$ and $\frac{\partial u}{\partial t}$ when $r = 2$, $s = 1$, and $t = 0$.

3. [15 pts] Find an equation for the tangent plane at the point $(1, -1, 1)$ to the surface

$$x^4 + y^4 - z^4 = x^2 y^2 z^2.$$

4. [15 pts] Find the absolute maximum and minimum values of $f(x, y) = x^2 + y^2 + 2x - 4y$ on the closed triangular region with vertices at $(0, 0)$, $(3, 0)$ and $(0, 3)$.

5. [15 pts] Find the point on the plane $x + 2y + 3z = 28$ that is closest to the origin.

6. [10+5 pts] Consider the function $f(x, y) = 5xe^y - x^2y$.

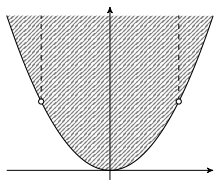
- (a) Find the rate of change of f at the point $(1, 0)$ in the direction of the vector $\mathbf{i} + 3\mathbf{j}$.
(b) Find the maximum value of the directional derivative of f at the point $(1, 0)$.

1. i. The correct choice is “it does not exist”.

ii. The correct choice is $2x + y - 1$.

iii. The correct choice is $\frac{y^w}{e^w - xy^w \ln y}$.

iv. The correct choice is $\frac{-y^6}{4(1 + xy^3)^{3/2}}$.



v. The correct choice is

2. Using the chain rule we obtain

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial r} = \frac{2x}{2\sqrt{x^2 + y^2}} \cdot 1 + \frac{2y}{2\sqrt{x^2 + y^2}} \cdot \sin t,$$

and

$$\frac{\partial u}{\partial t} = \frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial y} \frac{\partial y}{\partial t} = \frac{2x}{2\sqrt{x^2 + y^2}} \cdot (-s \sin t) + \frac{2y}{2\sqrt{x^2 + y^2}} \cdot r \cos t.$$

When $r = 2, s = 1, t = 0$ we have $x = 3$ and $y = 1$. Hence $\frac{\partial u}{\partial r} = \frac{3}{\sqrt{10}}$ and $\frac{\partial u}{\partial t} = \frac{2}{\sqrt{10}}$.

3. The given surface is a level surface of the function $F(x, y, z) = x^4 + y^4 - z^4 - x^2 y^2 z^2$, thus $\nabla F(1, -1, 1)$ is normal to the desired tangent plane. On the other hand,

$$\nabla F = \langle 4x^3 - 2xy^2z^2, 4y^3 - 2x^2yz^2, -4z^3 - 2x^2y^2z \rangle \implies \nabla F(1, -1, 1) = \langle 2, -2, -6 \rangle.$$

Hence the equation of the wanted tangent plane is

$$2(x - 1) - 2(y + 1) - 6(z - 1) = 0.$$

4. Let T denote the given triangular region. The absolute extrema of f on T occur at a critical point in the interior of T , or at a point on the boundary of T . The partial derivatives $f_x = 2x + 2$ and $f_y = 2y - 4$ are always defined, and $f_x = f_y = 0 \iff (x, y) = (-1, 2)$. Thus $(-1, 2)$ is the unique critical point of f . Since $(-1, 2)$ lies outside of T , it is not considered in our discussion. We consider each side of the triangular boundary of T separately:

- On the line segment from $(0, 0)$ to $(3, 0)$ we have $y = 0$ and $x \in [0, 3]$. The restriction of f to this line segment is $g_1(x) = f(x, 0) = x^2 + 2x$. Since g_1 has no critical number in the interval $(0, 3)$, its extreme values are $g_1(0) = 0$ and $g_1(3) = 15$.
- On the line segment from $(0, 0)$ to $(0, 3)$ we have $x = 0$ and $y \in [0, 3]$. The restriction of f to this line segment is $g_2(y) = f(0, y) = y^2 - 4y$. Since g_2 has the critical number $2 \in (0, 3)$, its extreme values are among the values $g_2(0) = 0$, $g_2(2) = -4$, and $g_2(3) = -3$.
- On the line segment from $(0, 3)$ to $(3, 0)$ we have $y = 3 - x$ and $x \in [0, 3]$. The restriction of f to this line segment is $g_3(x) = f(x, 3 - x) = 2x^2 - 3$. Since g_3 has no critical number in the interval $(0, 3)$, its extreme values are $g_3(0) = -3$ and $g_3(3) = 15$.

Therefore, the maximum and minimum values of f on the region T are $f(3, 0) = 15$ and $f(0, 2) = -4$ respectively.

5. Let $g(x, y, z) = x + 2y + 3z$. The desired point on the plane $g = 28$ minimizes the function $f(x, y, z) = x^2 + y^2 + z^2$ representing the square of distance to the origin. Thus by the method of Lagrange multipliers, this point is a solution of the system

$$\begin{cases} \nabla f = \lambda \nabla g \\ g = 28 \end{cases} \iff \begin{cases} 2x = \lambda \cdot 1 \\ 2y = \lambda \cdot 2 \\ 2z = \lambda \cdot 3 \\ x + 2y + 3z = 28. \end{cases}$$

Solving for x, y, z in terms of λ and substituting in the last equation gives

$$\frac{\lambda}{2} + 2\lambda + \frac{9\lambda}{2} = 28 \iff \lambda = 4 \iff (x, y, z) = (2, 4, 6).$$

We conclude that $(2, 4, 6)$ is the point on the plane $x + 2y + 3z = 28$ that is closest to the origin.

6. (a) The vector $u = \frac{\mathbf{i} + 3\mathbf{j}}{\sqrt{10}}$ is unit and in the same direction as the given vector. On the other hand,

$$\nabla f = \langle 5e^y - 2xy, 5xe^y - x^2 \rangle \implies \nabla f(1, 0) = \langle 5, 4 \rangle.$$

Hence the wanted rate of change is $D_{\mathbf{u}}f(1, 0) = \nabla f(1, 0) \cdot \mathbf{u} = \frac{17}{\sqrt{10}}$.

- (b) The maximum value of the directional derivative of f at the point $(1, 0)$ is $|\nabla f(1, 0)| = \sqrt{41}$.